

Name: DARISY SREETJRoll No: EE 25 BTCH 11018**Instructions**

1. The exam duration is 1 hour, starting at 12:10. Ensure that you submit both the question paper and the answer book by 13:10. Please be aware that **late submissions** may incur **penalties**.
2. Read all questions carefully. Misinterpretations will **not** be considered. It is essential to write your solutions clearly and legibly in the answer book. In a previous quiz, a few students submitted answers that were not clearly written. Going forward, such responses will **not** be evaluated.
3. Any form of unfair practice will result in an **F grade** and will be reported to the disciplinary committee.

1. (4 points) Derive the set of differential equations that describe the time-domain behavior of the higher-order circuit shown in Fig. 1. Assume that the circuit is initially at rest. You are free to choose any circuit analysis technique to derive the equations. Clearly mention the circuit analysis technique you are using. Make sure that the equations

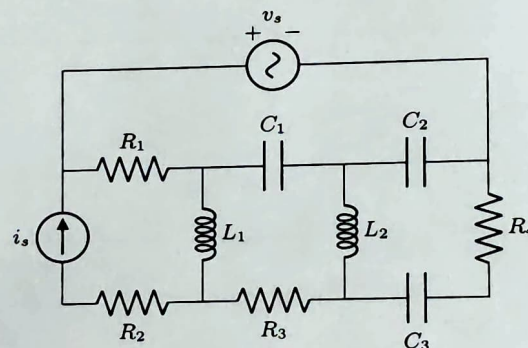


Fig. 1: Higher-order circuit for Question 1

are in differential form (i.e., involving derivatives of variables with respect to time).

2. (6 points) Derive the Thevenin equivalent circuit seen from the terminals a-b of the circuit shown in Fig. 2. Clearly indicate all the steps involved in your derivation.
 - (a) (2 points) Determine the Thevenin equivalent voltage V_{TH} .
 - (b) (2 points) Determine the Thevenin equivalent resistance R_{TH} .
 - (c) (1 point) Determine the value of load resistance R_L that would maximize the power delivered to the load. Also, calculate the maximum power delivered to the load.
 - (d) (1 point) Compute the voltage across the load under maximum power transfer conditions.

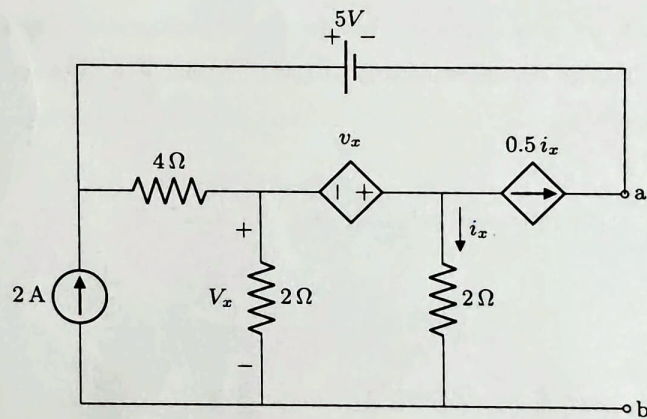


Fig. 2: Circuit for Question 2